

Zachary Bauer

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Education

University of Michigan, Ann Arbor, MI

Bachelor of Science in Engineering, Computer Engineering, Cum Laude

Aug 2020 - May 2024

Relevant Courses: Advanced Embedded Systems, Applied GPU Programming, Analog Circuits

GPA: 3.458

Masters of Science, Electrical and Computer Engineering

Aug 2024 - Present

Relevant Courses: Monolithic Amplifier Design, Discrete Event Systems, Embedded Control Systems

GPA: 3.890

Experience

Graduate Researcher

May 2024 – Present

Interactive Sensing Computing Lab, University of Michigan, Ann Arbor, MI

- Developed tool-chain for taking Altium files and turning them into a format that can be laser cut onto copper-clad FR4 with a fiber laser, and used tool-chain to design and fabricate multilayer PCBs
- Built amplifier circuits to explore ultrasonic wave generation for sound transmission
- Created 4-layer development PCB kit for testing the ESP recording and cataloging capabilities
- Developed Rigid/Flex PCB for use in phone case for monitoring project
- Modeling microphone array for beamforming in Matlab
- Designing the PCB for the microphone array
- Designing wrist-mounted wearable for privacy aware sensing and recording

Research Intern

May 2025 – July 2025

Fibarcode, Ann Arbor, MI

- Designed and built draw tower for polymer fiber creation
- Created control system for operation of the draw tower
- Coded interface for control system
- Created simulation tool to analyze different layer stacks of fiber in Lumerical

Graduate Student Instructor - Engineering Interactive Systems

Jan 2025 – April 2025

University of Michigan, Ann Arbor, MI

- Created lab content for and taught labs in laser cutting, 3D printing, circuits, basic machine learning, and basic mobile app development
- Provided support for group final semester project through office hours and weekly meetings with each team

Product Development Intern

May 2023 – July 2023

Ford Motor Company, Dearborn, MI

- Worked for 12 weeks on Battery Management Hardware Team
- Developed device for testing the next generation BMS for accuracy

Computer Consultant

May 2022 – August 2022

University of Michigan, Ann Arbor, MI

- Assisted with integration of smart boards into university classrooms and faculty training
- Responsible for troubleshooting and smart board tech support
- Worked on AR/VR development for alternative teaching methods

Skills

- **Software:** Cadence, Altium, Kicad, Fusion360, LTspice, Matlab
- **Programming Languages:** C/C++, ARM assembly, Verilog HDL, Matlab, Python, Bash, HTML
- **Debugging Tools:** GDB and LLDB, Valgrind, and Perf

Projects

Smart Scoreboard

September - December 2023

- Second Senior Capstone project for Advanced Embedded Systems course
- Designed and assembled a PCB for wearable application
- Created protocol for sending commands to the scoreboard wirelessly

Skin Touch Tracking Device

February - April 2023

- Senior Capstone project for Engineering Interactive Systems course
- Built oscillator that created a 60 MHz electrical signal that was transmitted through the skin
- The device was able to detect and localize a person's finger when they touched their arm